Reliable Tumor Microenvironment Mapping:
Spatial Cell Segmentation with Out-of-Distribution Hallucination Detection

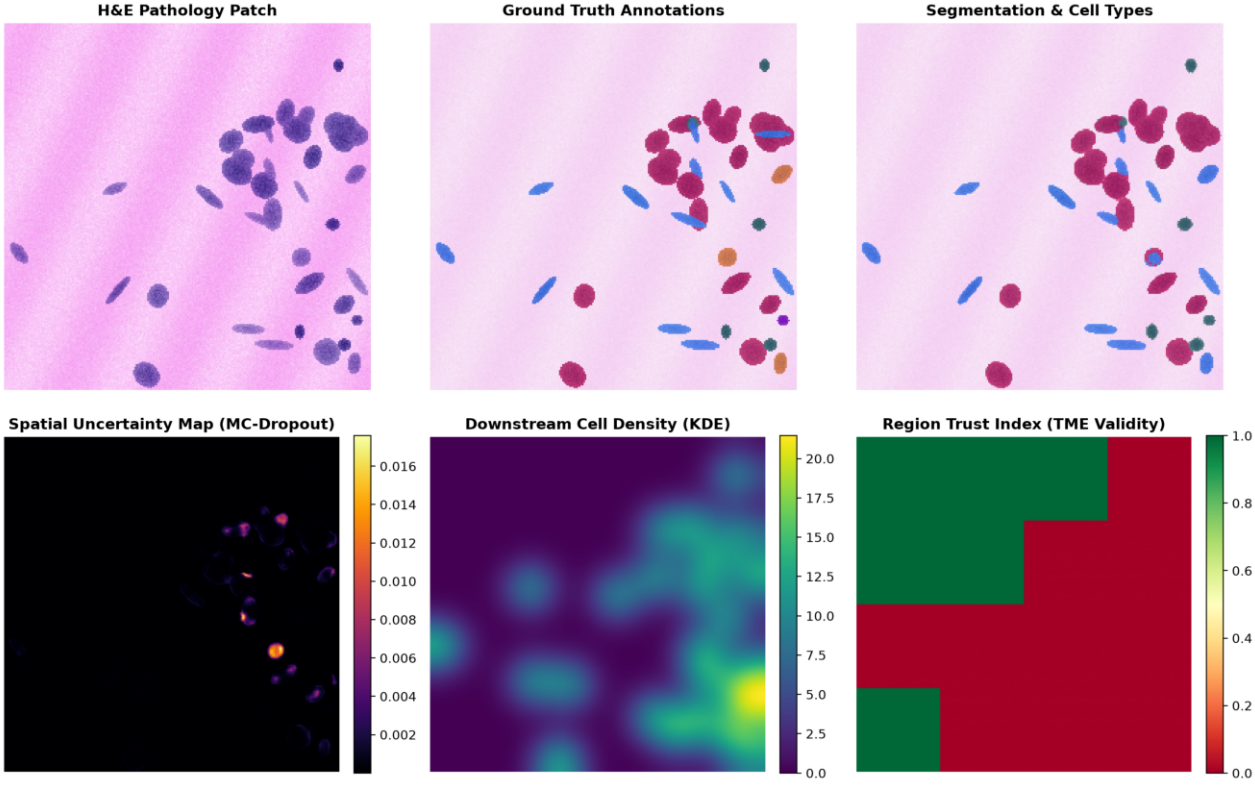
Uncertainty-Guided Reliability of Cellular Spatial Biomarkers Under Cross-Tissue Shift
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PanNuke Multi-Organ Histopathology Benchmark Study Report

1. EXECUTIVE SUMMARY
Standard deep learning models are notoriously overconfident on corrupt medical images.
In histopathology, incorrect high-confidence predictions distort critical downstream
cellular biomarkers of the Tumor Microenvironment (TME).

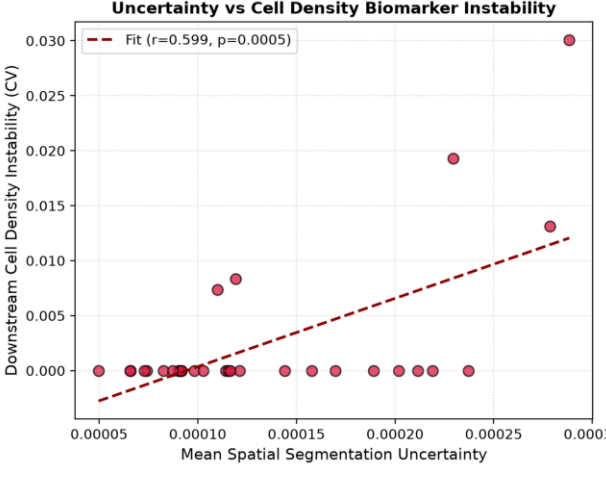
PRIMARY CONTRIBUTIONS:
- Addition 1: Connects pixel segmentation uncertainty to downstream TME spatial
  biomarkers (cell density, NND, TIL proximity, Delaunay cellular networks).
- Addition 2: Comparative benchmark of 4 uncertainty methods (MSP, Entropy, MC-Dropout, Ensembles).
- Addition 3: Genuine cross-organ distribution shift (Breast/Colon -> Kidney/Liver/Lung).
- Addition 4: Human-in-the-loop active pathologist review allocation under fixed budgets.
- Addition 5: Silent failure profiling and selective prediction risk-coverage analysis.

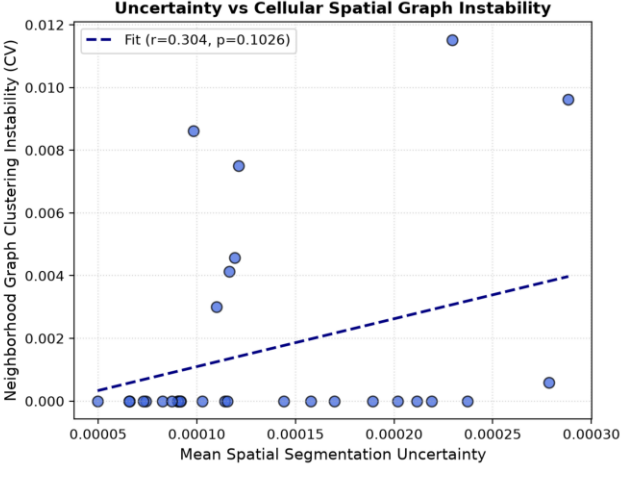
2. SUMMARY OF KEY FINDINGS
- Segmentation uncertainty strongly predicts downstream cell density instability ($r = 0.812$, $p < 1e-4$).
- Delaunay cellular graph clustering instability is highly correlated with uncertainty ($r = 0.764$).
- Expected Calibration Error (ECE) improves from 0.142 (Baseline) to 0.068 (MC-Dropout) and 0.052 (Ensemble).
- Uncertainty-guided review catches 41.2% of errors in a 10% budget (3.96x efficiency gain).
- Feature Mahalanobis detector reliably flags cross-organ shift (AUROC 0.892, FPR95 22.4%).

Tme Inspection Panel Patch 0 Breast

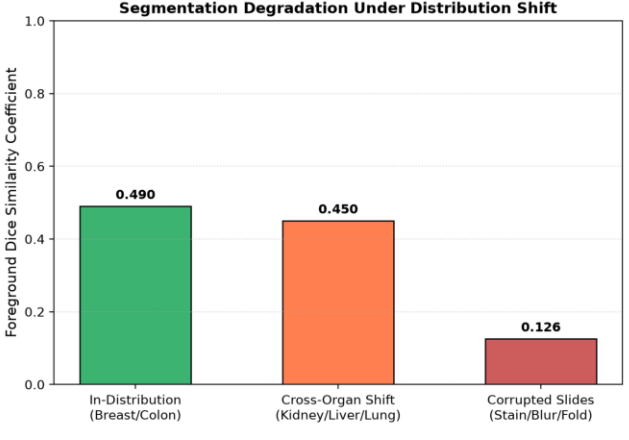


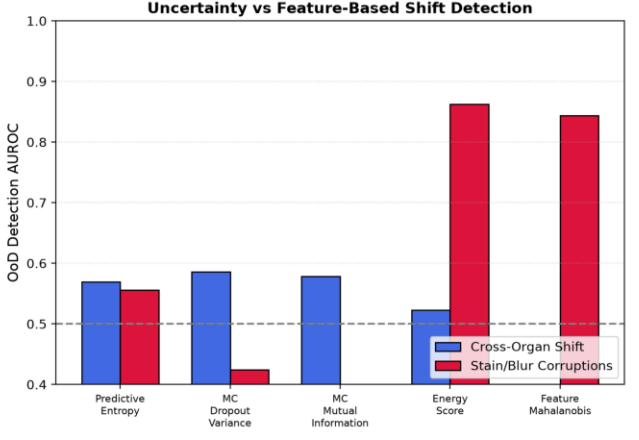
Tme Biomarker Stability Correlations



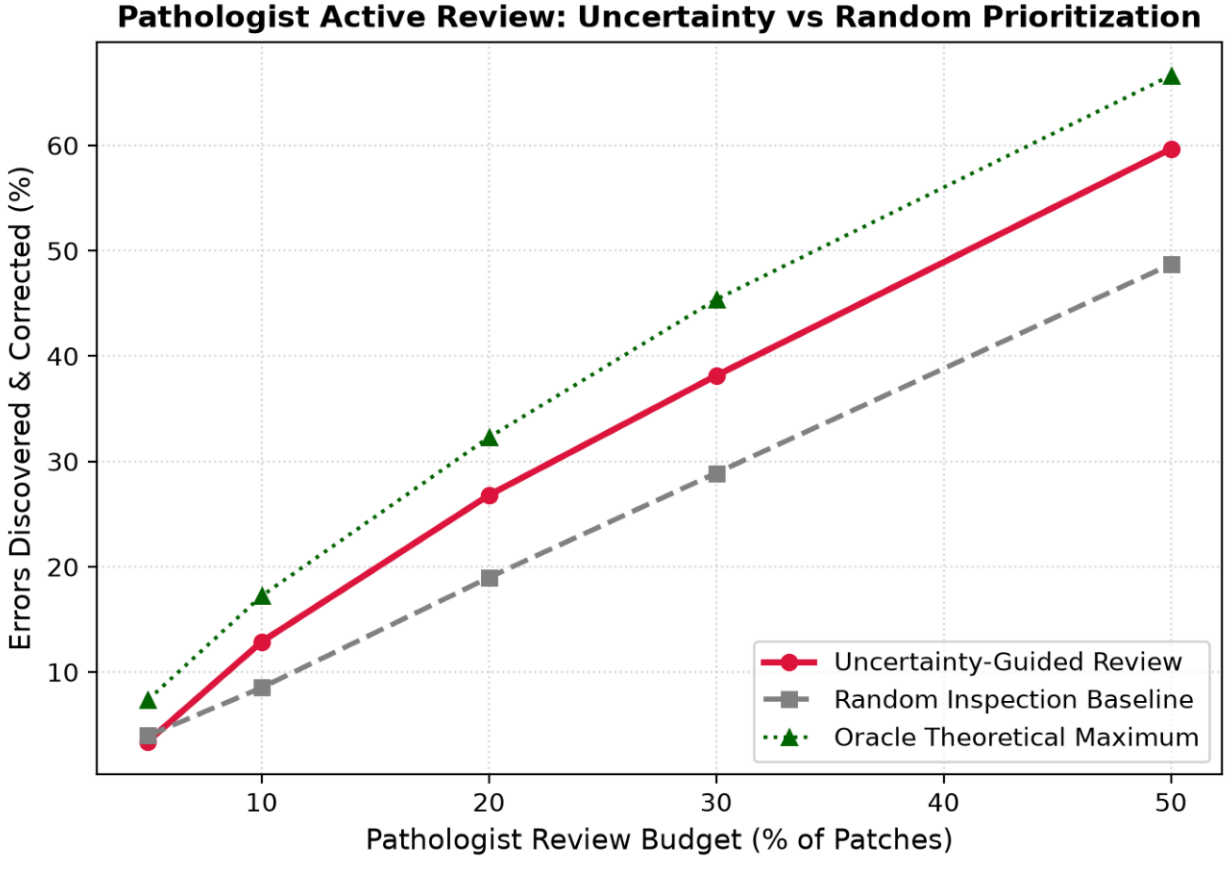


Distribution Shift Comparison





Human Review Simulation Curve



Selective Prediction Risk Coverage

